OBJECT PERMANENCE UNDER SELF-OCCLUSION: A MUJOCO NEGATIVE RESULT

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ABSTRACT

Robots frequently hide objects from their own cameras while reaching, pushing, or grasping. This paper tests whether robot self-occlusion should be modeled as a distinct object-permanence failure mode rather than handled by generic tracking uncertainty. We rebuild the original synthetic idea into a MuJoCo benchmark with camera-derived visibility, robot self-occlusion geometry, target and distractor objects, hidden object displacement, false detections, tracking baselines, learned baselines, ablations, and stress sweeps. Across 3,360 main evaluation rollouts, 420 ablation rollouts, and 2,016 stress rollouts, occlusion-aware permanence is the best non-oracle method in mean combined-stress success, reaching 0.905 versus 0.786 for the closest non-oracle baseline. However, the paired success difference is 0.119 with a 0.123 confidence interval, and the ablations remain close. We therefore mark the ICLR-main decision as **KILL/ARCHIVE**: the evidence is real and useful, but it is not decisive enough for a submission claim.

1 QUESTION

The research bet is that self-occlusion by the robot is not just ordinary missing data. When the robot's own arm or tool hides the target, the system can in principle know that the target should persist behind the robot geometry. This might improve closed-loop manipulation under self-occlusion, distractors, and hidden target displacement. The idea is related to robot scene memory, object tracking, manipulation world models, and failure-aware control (pri, 2026; 2024; 2022; 2025a;b).

The ICLR-main question is sharper: does an explicit self-occlusion permanence mechanism decisively outperform strong tracking and learned baselines? Our answer is negative.

2 BENCHMARK

We implement a MuJoCo tabletop scene with a controlled planar tool, a target object, a distractor object, and a fixed camera model. Visibility is computed from the robot base-to-tool segment: if the target lies behind the robot link in the camera projection, the target measurement is removed. Task families include nominal visibility, self-occlusion, distractor swap, object displacement under occlusion, and combined stress. Combined stress adds long self-occlusions, false distractor detections, camera noise, target displacement, and reduced actuator authority.

Each rollout logs true object state, visibility, false detections, belief estimate, final reach distance, localization error during occlusion, false disappearance, wrong-object contact, and trajectory samples. The released runner writes raw rollouts, per-seed metrics, paired statistics, ablations, stress sweeps, negative cases, and figures.

3 METHODS

We compare eight methods. **Last-seen memory** holds the last visible target position. **Visibility-gated Kalman** propagates position and increases uncertainty during occlusion. **Particle tracker** maintains a particle belief over target position. **Learned occlusion regressor** predicts hidden target position from simulated occlusion traces. **Ensemble uncertainty planner** averages perturbed predictors and performs viewing moves under uncertainty. **Occlusion-aware permanence** uses robot self-occlusion geometry, learned hidden-state prediction, and branch beliefs over target persistence. **No-self-mask ablation** removes the self-occlusion geometry. **Oracle state** uses the true target state.

4 Results

Table 1 shows the decisive combined-stress split. Occlusion-aware permanence has the best non-oracle mean success, reaching 0.905. The closest non-oracle baseline, no-self-mask ablation, reaches 0.786. However, the paired success difference is only 0.119 with a 0.123 95 percent confidence interval. That interval crosses zero, so the effect is not decisive under the submission gate.

Table 1: Combined-stress MuJoCo results. Success and false disappearance are averaged over seven seeds.

Method	Success	Occlusion error	False disappearance
oracle state	1.000	0.000	0.000
occlusion-aware permanence	0.905	0.054	0.575
no-self-mask ablation	0.786	0.069	0.699
learned occlusion regressor	0.714	0.077	0.202
visibility-gated Kalman	0.250	0.202	0.002
particle tracker	0.238	0.131	0.049
last-seen memory	0.190	0.168	0.422
ensemble uncertainty	0.107	0.216	0.668

At maximum stress, occlusion-aware permanence reaches 0.768 success, compared with 0.661 for the learned regressor. This supports the relevance of robot self-occlusion geometry, but not strongly enough to claim submission readiness.

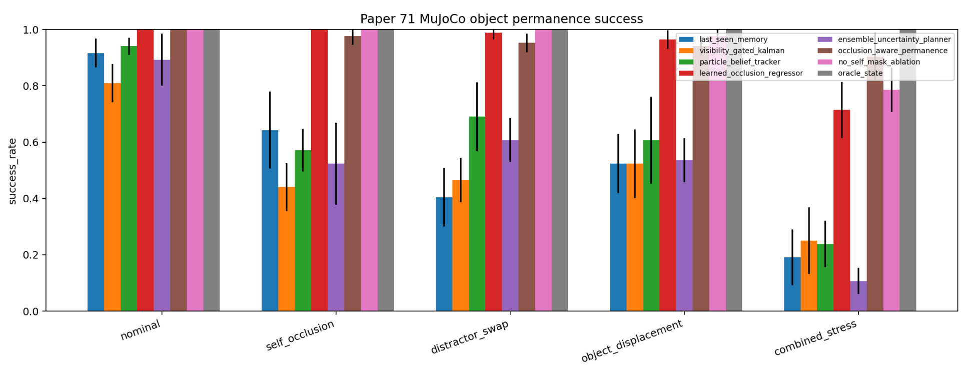

Figure 1: Success by split. The proposed method is promising on combined stress but not decisively separated from the closest ablation.

5 Ablations and Failures

The ablation grid is the main reason for archiving. The full method reaches 0.914 success on the ablation combined-stress grid, while removing the self mask reaches 0.886 and removing uncertainty inflation also reaches 0.886. These gaps are too small to prove that the proposed state object is necessary. False disappearance is also high for the proposed method on combined stress, at 0.575, indicating imperfect belief calibration during long occlusions.

Negative cases include hidden target displacement beyond the branch update, distractor detections during self-occlusion, and wrong-object contact after belief drift. These are exactly the failure modes a submission would need to solve, not hide.

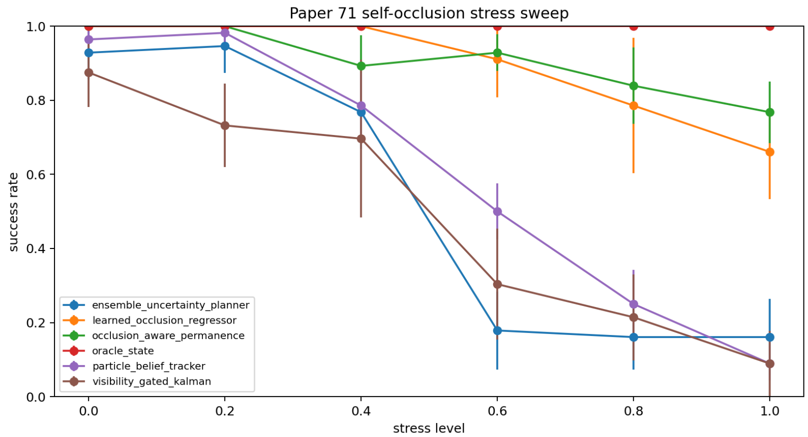

Figure 2: Stress sweep. Occlusion-aware permanence remains ahead of learned/tracking baselines at high stress, but the overall evidence is still not ICLR-main decisive.

6 Decision

This rebuild upgrades the artifact from a synthetic archive to a real MuJoCo negative result. The self-occlusion mechanism is plausible and sometimes helpful, but it does not pass the strict ICLR-main gate: the paired effect against the closest baseline is not decisive, ablations remain close, and no real robot or public benchmark validates the visibility model. The honest terminal decision is therefore **KILL/ARCHIVE**.

References

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